Netlink Intern & Trainee Lifecycle Management Platform

Product Requirements Document — User Workflows, Core Problem Solved & Future Roadmap

1. Platform Purpose & What the Application is Made On

The **Netlink Intern/Trainee Platform** is an enterprise web application designed to govern, automate, and track the end-to-end journey of student interns and graduate trainees—from recruitment, biometric enrollment, and geofenced attendance, to task assignments, mentoring, fee tracking, institutional college cohorts, and graduation clearance.

Tier / Technology	Core Frameworks & Libraries	Purpose in the Application
Frontend UI	React 18 + TypeScript + Tailwind CSS + Vite	Fast, responsive single-page dashboards (Student, Mentor, HR, Admin), live webcam video capture for face match, GPS query.
Backend API	Python 3.11/3.14 + FastAPI + Pydantic V2	High-speed REST API endpoints, server-side RBAC validation, Haversine geofence boundary calculations, daily attendance engine.
Database & Storage	PostgreSQL via Supabase + SQLAlchemy ORM	Relational ACID database for 27 normalized tables, cloud object storage for deliverables and generated letters, SQLite local fallback.
Security & Auth	Supabase Auth (JWT) + Biometric Vision Layer	Bearer token verification, 512-dimensional facial embedding vectors (no raw photos stored), hardware device fingerprint binding.

2. What Our Application Solves (Problem vs. Solution)

Before this platform, managing dozens of training cohorts, hundreds of college trainees, and multiple corporate locations resulted in operational bottlenecks, proxy attendance, delayed grading, and compliance risks.

Key Operational Challenge	The Problem in Traditional Systems	How the Netlink Platform Solves It
Attendance Fraud & Proxy Punches	Trainees share credentials or use fake GPS apps to mark attendance while absent.	**Zero-Trust Verification:** Strict 100m server-side Haversine geofence + live facial matching (0.82 score) + primary device fingerprint lock.
Fragmented Trainee Management	Syllabus, attendance, tasks, and grades live in scattered spreadsheets and chat groups.	**Unified Single Source of Truth:** Centralized portal for syllabus topics, submissions, deadlines, and skill evaluations.
Manual Exit & No-Dues Clearance	Graduating trainees run between mentors, lab coordinators, and accounts to collect physical signatures.	**Automated 4-Stage Digital Clearance:** Sequential online sign-off (Mentor -> Assets -> Accounts -> HR) with automated certificates.
Institutional College Tracking	Difficulty tracking MoU terms, college coordinators, and cohort fees across partner universities.	**Dedicated Institutional Module:** Tracks MoU dates, college student rosters, batch performance, and installment fee payments.
Compliance & Audit Gaps	No verifiable trail for manual attendance overrides, status changes, or document issuance.	**Immutable Security Audit Logs:** Every administrative action, override, and role change is permanently logged with timestamps.

3. Users & Their Working (Roles & Responsibilities — Part 1)

The platform organizes all operations around four core user roles. Each role has a dedicated dashboard, strict data privacy boundaries, and specific day-to-day workflows.

3.1 Student / Trainee — Everyday Working & Functions

Who They Are: College interns, technical trainees, and apprentices participating in Netlink training tracks (e.g. Full Stack, Cloud, QA).
Privacy Boundary: Students can strictly only view and modify their own records. They have zero access to other students' data.

- **1. Biometric Onboarding & Device Binding:** Upon first login, the trainee reads and grants explicit GDPR/DPDP biometric consent. The system captures their live facial embedding template and binds their primary laptop/mobile device fingerprint.
- **2. Daily Geofenced Attendance Punching:** To punch in, the trainee opens the app on-campus. The camera captures their live face and GPS coordinates. If within 100 meters of the venue and face matches (>= 0.82), the punch is recorded. Cumulative 5+ hours (300 mins) logs a 'PRESENT' status.
- **3. Task Submissions & Deliverables:** Students review active assignments, deadlines, and requirements. Deliverables (GitHub repository links, live demo URLs, or files) are submitted directly through the portal.
- **4. Curriculum & Skill Radar Monitoring:** Students mark syllabus topics as 'In Progress' or 'Completed' and view interactive Radar Charts illustrating their technical and soft skill proficiency.
- **5. Regularization & Leave Requests:** If a punch is missed due to official travel or technical fault, trainees submit a regularization ticket with reason and times for mentor approval.
- **6. No-Dues Clearance & Letters:** At program completion, trainees initiate an exit request, track departmental clearance approvals, and download their official Offer Letters and Completion Certificates.

3.2 Mentor / Technical Trainer — Everyday Working & Functions

Who They Are: Senior engineers, instructors, and technical leads responsible for teaching and evaluating specific batches.
Access Scope: Mentors can only view and manage students assigned to their specific cohorts (batch.mentor_id == current_user.id).

- **1. Batch Overview & Roster Tracking:** Mentors view their assigned batches, student rosters, daily attendance percentages, and identify students falling behind.
- **2. Task Assignment & Deadlines:** Mentors publish technical assignments, assign maximum point values, map tasks to curriculum skills, and set strict submission deadlines.
- **3. Submission Grading & Rubric Scoring:** Mentors inspect student code repositories, review demo links, assign scores (1–5 scale), and provide detailed qualitative feedback.
- **4. 1-on-1 Qualitative Reviews:** Conducts periodic reviews evaluating technical aptitude, problem solving, teamwork, and punctuality with actionable improvement suggestions.
- **5. Attendance Regularization Approval:** Mentors review and approve/reject punch correction tickets submitted by trainees in their batch.
- **6. Technical Exit Sign-off:** Signs off on project code handoff and documentation during the trainee's graduation No-Dues clearance process.
- **7. Trainer Leave & Substitute Scheduling:** Mentors apply for personal leaves through the portal, designating substitute trainers to maintain uninterrupted batch classes.

3. Users & Their Working (Roles & Responsibilities — Part 2)

Administrative roles maintain program-wide oversight, institutional partnerships, governance, and tamper-proof security.

3.3 HR & Program Manager — Everyday Working & Functions

Who They Are: Talent operations teams, HR managers, and training coordinators overseeing institutional cohorts.
Access Scope: Program-wide operational authority over users, batches, geofences, finances, and official documentation.

- **1. Program & Batch Architecture:** Creates training programs (Tech Master, Short Duration, Internship) and schedules cohort batches with start/end dates, mentor assignments, and capacities.
- **2. Trainee Onboarding & Batch Allocation:** Enrolls students, verifies college documentation, maps trainees to designated batches, and handles cohort transfers.
- **3. Geofence & Campus Venue Configuration:** Defines authorized physical training locations by latitude/longitude center points and radius thresholds (default 100m).
- **4. Hardware Device Swap Approvals:** Reviews student device registration requests. If a student loses or replaces their registered phone/laptop, HR approves the new device fingerprint.
- **5. Institutional College Partnership Management:** Manages college MoUs, university coordinator contacts, college cohort rosters, and institutional performance metrics.
- **6. Fee Accounting & Installment Receipts:** Records student and college fee payments, tracks paid vs. pending balances, and generates official payment receipts.
- **7. Official Letter Generation Engine:** Generates 1-click official documents: Offer Letters, Internship Completion Letters, Bonafide Letters, Letters of Recommendation, and Warning Letters with digital verification.
- **8. Final Exit Clearance & Certificate Issuance:** Executes the final HR sign-off on No-Dues clearance once mentor, lab, and accounts have approved, triggering automatic completion certificate generation.
- **9. Manual Attendance Overrides:** Authorized to correct attendance records for approved official duty, requiring an explicit audit explanation.

3.4 Super Administrator — Everyday Working & Functions

Who They Are: System administrators and IT security officers possessing highest-level platform governance.
Access Scope: Global read/write access across all system entities, security ledgers, and server configurations.

- **1. User Provisioning & RBAC Management:** Provisions staff accounts, activates/deactivates users, and modifies role claims (STUDENT, MENTOR, HR, SUPER_ADMIN).
- **2. Immutable Security Audit Log Monitoring:** Inspects the permanent audit trail recording every administrative override, role promotion, device approval, and failed login.
- **3. Global Security & Accuracy Thresholds:** Configures system-wide parameters such as facial recognition cosine threshold (0.82) and GPS accuracy cutoffs (<= 30m).
- **4. System Health & Infrastructure Supervision:** Monitors API latency, database connection pools, Supabase cloud storage, and background task health.

Role Access & Permission Matrix at a Glance:

System Function	Student	Mentor	HR Manager	Super Admin
Punch Attendance (GPS + Face)	Own Only	No	No	No
Approve Attendance Regularization	No	Assigned Batch	Global	Global
Create & Grade Task Assignments	Submit Only	Assigned Batch	Global	Global
Manage Programs, Batches & Fees	No	View Only	Full Manage	Full Manage
Approve Devices & Generate Letters	Request Only	No	Full Access	Full Access
User Provisioning & Security Audit Logs	No	No	View Limited	Full Hegemony

4. What the Application Will Have in the Future (Roadmap)

While Phase 1 (Core Lifecycle) and Phase 2 (Enterprise Operations Suite) are currently deployed and active, the future development roadmap focuses on AI intelligence, placement ecosystems, and cross-platform mobility.

Planned Capability / Milestone	Architecture & Integration Scope	Target Impact & Benefits
AI Advisory Assistant (Groq + LangChain)	Fast inference using Groq LPU models (Llama-3-70B). Analyzes student task submissions, identifies skill gaps, drafts personalized mentor feedback, and recommends learning resources.	Saves 70% of mentor feedback drafting time. *Strict Guardrail: AI is strictly advisory; never makes final grading, hiring, or dismissal decisions.*
Corporate Placement & Job Matching Portal	Internal career portal matching high-performing trainees to open client project roles at Netlink based on evaluated skill radar scores and attendance reliability.	Accelerates post-training talent absorption from weeks to hours, giving delivery managers direct access to verified candidate rankings.
Cross-Platform Mobile App (React Native / Flutter)	Native iOS and Android mobile applications featuring background geofencing beacons, biometric Secure Enclave key storage, and offline punch caching with cryptographic sync.	Enables zero-friction attendance check-in on smartphones even in low-connectivity training labs, with automatic sync upon reconnect.
Enterprise HRMS & LMS Bi-Directional Connectors	Pre-built REST and webhook connectors for Workday, SAP SuccessFactors, BambooHR, and GitHub Classroom.	Eliminates double data entry; graduated trainees automatically transition into full-time employee HRMS records.
Predictive Dropout & Early Intervention Analytics	Machine learning models analyzing leading indicators (attendance drops, late task submissions, feedback scores) to flag at-risk trainees early.	Enables proactive HR and mentor intervention before a trainee drops out, improving program completion rates by over 25%.

CORE TAKEAWAY: The Netlink Platform transforms a fragmented, vulnerable trainee onboarding process into a **zero-trust, transparent, and auditable talent lifecycle ecosystem**. By bridging biometric precision, curriculum tracking, institutional college partnerships, and future AI advisory capabilities, the platform guarantees operational excellence for students, trainers, and enterprise leadership alike.

5. Document Governance & Stakeholder Approval

Stakeholder Role	Designated Name	Approval Status	Date
Lead Product Manager	Talent Operations Lead	**APPROVED (Digital Sign-off)**	12-Oct-2026
Lead Solutions Architect	Platform Engineering Lead	**APPROVED (Digital Sign-off)**	12-Oct-2026
Head of HR & Training	VP Human Capital	**APPROVED (Digital Sign-off)**	13-Oct-2026
Information Security Officer	Chief Security Officer	**APPROVED (Tier-1 Biometrics)**	13-Oct-2026